

# R1a — Participant-level Circadian Curve Features (Garmin)

Plan-language summary · Datenspende sex-imputation program

**Status:** run 2026-09-21, commits 4868482 + a494e35 . 30 repeats × 5 arms, wall 447 s, no rescan (raw epochs untouched). **Standing decision:** vendor-circularity audit **declined** (2026-09-21); ch3000-only is the convention for feature-side arms, not a deferral.

## Cohort & label

- s3 (Garmin) strict-coverage model union, **n = 3,848**; 70/15/15 → 2692/578/578 per repeat.
- Label = recorded salutation,  $y=1 \Leftrightarrow$  female-form (2464/1384 = 64.0% prevalence).
- Splits = the frozen FS-phase participant splits: `SeedSequence([20260919, 3, r])`, per-class permutation, test→val→train. **Paired with every prior experiment** (FS, FE, age, dayscale).

## Input

`artifacts_v2/epoch_hours.parquet`, filtered `channel = 3000`  $\wedge$  `user`  $\in$  `cohort` → 92,286 (user, hour-of-day) cells, coverage 1.0000. Schema (`user`, `channel`, `hour`  $\in$  {0..23}, `n`, `vsum`, `vsumsq`): per-user sums of all valid minute-observations grouped by **local clock hour** over the whole recording. Zero-cost source — the discovery that made R1a free.

## Features (35 per user)

Per user, hourly mean and weight over the recording:

$$\bar{h}_k = \frac{\sum \mathbf{vsum}}{\sum \mathbf{n}}, \quad n_k = \sum \mathbf{n}, \quad k = 0, \dots, 23$$

**Curve block (11).** Weighted cosinor, 3 harmonics, weighted least squares with weights  $W = \text{diag}(n_k)$ :

$$\widehat{\text{HR}}(k) = M + \sum_{j=1}^3 \left[ a_j \cos\left(\frac{2\pi jk}{24}\right) + b_j \sin\left(\frac{2\pi jk}{24}\right) \right], \quad \hat{\beta} = (X^\top W X)^{-1} X^\top W \bar{h}$$

solved via `lstsq` (min-norm under rank deficiency). Emits: mesor  $M$ ; harmonic amplitudes  $A_j = \sqrt{a_j^2 + b_j^2}$  (24/12/8-h); acrophase

$$\text{acro} = \arg \max_{t \in [0, 24)} \widehat{\text{HR}}(t)$$

evaluated on a 0.25-h grid (phase-convention-free), and goodness-of-fit

$$\text{resid\_sd} = \sqrt{\frac{\sum_k n_k (\bar{h}_k - \widehat{\text{HR}}(k))^2}{\sum_k n_k}}$$

plus the curve summaries

$$\text{night\_mean} = \frac{\sum_{k < 6} n_k \bar{h}_k}{\sum_{k < 6} n_k}, \quad \text{contrast} = \text{wmean}(\bar{h}, 8..19) - \text{wmean}(\bar{h}, 0..5), \quad p_k \propto (\bar{h}_k - \min \bar{h})^+, \quad H = - \sum_k p_k \ln p_k$$

with `morning_slope` = weighted-LS slope of  $\bar{h}_k$  on  $k$  over  $k \in \{5, \dots, 9\}$  (bpm/h) and `range` =  $\max \bar{h} - \min \bar{h}$  over worn hours.

**Raw profile (24).** `hour_h0..h23` =  $\bar{h}_k$ , NaN where  $n_k = 0$  (structural; median-imputed downstream).

**Wear-partialled variant** (per repeat, OLS fit on train users only):

$$f_{\text{resid}} = f - \left[ \hat{\gamma}_0 + \sum_{k=0}^{22} \hat{\gamma}_k s_k + \hat{\gamma}_{23} \log n_{\text{total}} \right], \quad s_k = \frac{n_k}{\sum_k n_k}$$

i.e. each curve feature linearly residualised on its own hour-of-day coverage profile; residuals applied to all users (train-fit coefficients).

## Protocol

- Arms (5):** BASE (FS-A1 replica), `R1a_only`, `BASE@R1a`, `R1a_resid_only`, `BASE@R1a_resid`.
- Hygiene:** SQPreprocessor on train (median impute + one-hot + indicators) → A1: drop `__missing`, caret nzv (dominant >95%  $\wedge$  unique/n < 10%). 400 cols BASE → 435 combined (no R1a column dropped).
- Model:** RF G1 @ 100 trees — `max_features=0.4`, `min_samples_leaf=10`, `class_weight=balanced_subsample`; seed `SeedSequence([20260919, 3, r, 5])` — the same stream as FS/age, which is what makes BASE bit-exact.

## Inference

Primary family: s3 `all`, two comparisons vs BASE, paired per repeat:

$$\Delta_r = \text{AUROC}(\text{arm}, r) - \text{AUROC}(\text{BASE}, r), \quad r = 0, \dots, 29$$
$$\bar{\Delta} \pm t_{0.9875, 29} s / \sqrt{30}$$

Two 97.5% CIs  $\Rightarrow$  Bonferroni-simultaneous 95% family. Detection floor  $\approx$  0.003 AUROC (MC SE  $\approx$  0.0015).

Results

**Gate:** BASE = frozen FS A1, max  $|\Delta| = 0\text{e}+00$  over 30/30 repeats.

| arm            | test AUROC (mean $\pm$ SD)            |
|----------------|---------------------------------------|
| BASE           | 0.7434 $\pm$ 0.0155                   |
| R1a_only       | 0.6921 $\pm$ 0.0200                   |
| R1a_resid_only | 0.6885 $\pm$ 0.0157                   |
| BASE@R1a       | 0.7463 $\pm$ 0.0179                   |
| BASE@R1a_resid | <b>0.7470 <math>\pm</math> 0.0173</b> |

**Primary comparisons** (97.5% t-CI, Bonferroni-simultaneous 95% — both clear zero):

| comparison            | mean $\Delta$  | 97.5% CI           |
|-----------------------|----------------|--------------------|
| BASE@R1a – BASE       | +0.0029        | [+0.0004, +0.0054] |
| BASE@R1a_resid – BASE | <b>+0.0036</b> | [+0.0014, +0.0058] |

**Direction read** (class 0 = male-form salutation, class 1 = female-form):

| feature                    | class 0 (male) | class 1 (female)        |
|----------------------------|----------------|-------------------------|
| cosinor_acro_h (peak time) | 13.76 h        | 14.53 h (+0.77 h later) |
| curve_night_mean (00–05)   | 64.8 bpm       | 60.8 bpm                |
| hour_h9                    | 79.6 bpm       | 75.1 bpm                |
| hour_h20                   | 77.6 bpm       | 75.1 bpm                |
| cosinor_A3 (8-h amplitude) | 1.77           | 1.57                    |

Key findings

- **Both primary comparisons clear zero** — the first feature-side lever past the detection floor since FS-phase saturation; same order as the age counter (+0.0029).
- **Standalone density:** 35 curve features alone reach 0.6921, i.e.  $\frac{0.6921-0.5}{0.7434-0.5} = 79\%$  of the participant-level excess signal.
- **Shape, not coverage:** residualising the hour-of-day coverage profile out of each curve feature *before* addition raises the increment (+0.0036 vs +0.0029) — the added information is relative timing/form of the hourly profile, not wear volume. Coverage-partialling is a cheap denoiser worth carrying into R1b.
- **Direction is opposite to lab resting physiology** (women: higher resting HR, earlier phase): male curves run hotter at every clock hour and peak earlier. The classifier is likely reading activity load / cohort composition / wear behaviour, not textbook circadian dimorphism. Label is recorded salutation, not biological sex/gender.
- **Positioning:** the *trait* complement to dayscale’s *state* result. Participant level is near-saturated; the day level (base  $\approx$  0.65) has the headroom — R1b (per-day curves, weekday/weekend contrast, minute-scale dynamics, episodes) should be evaluated there first; its day-curve cache doubles as the D2 warm-start input.